Question Paper ID: 335054

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Course Code: CS372IA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

COMPUTER SCIENCE AND ENGINEERING

Parallel Architecture and GPU Programming

Time : 3 Hours

Instructions to the students

Maximum Marks : 100

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	The relation for Mean time between failures (MTBF) is determined as the sum of ____________________ and________________.	02	2	2
1.2	Examine the role of the Reorder Buffer (ROB) in speculative execution	02	2	4
1.3	What is the key bottleneck in a Symmetric Shared Memory (UMA) architecture?	02	2	4
1.4	State two synchronization primitives.	02	1	1
1.5	State two advantages of vector architecture	02	1	1
1.6	Differentiate between CUDA thread, block, and grid	02	2	2
1.7	What is load balancing with respect to parallel algorithm design.	02	2	2
1.8	Compare speculative decomposition with exploratory technique	02	2	2
1.9	Distinguish between copyin and copyout clauses of OpenACC	02	1	2
1.10	What is the role of a gang in OpenACC execution?	02	1	2

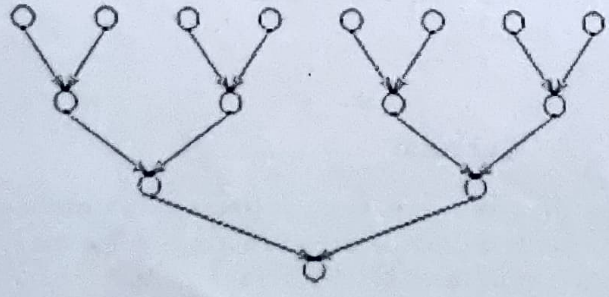
Part B

Question No	Question	M	CO	BT
2a	Define loop unrolling. For the given code snippet Show loop unrolled for four copies of the loop body. Schedule the unrolled loop for the pipeline with the latencies(give in table below). Eliminate any obviously redundant computations and do not reuse any of the	10	4	5

	registers. Determine the total clock cycles needed to execute the loop for both cases Loop: L.D F0,0(R1) ; ADD.D F4,F0,F2 ; S.D F4,0(R1) ; DADDUI R1,R1,#-8 ; BNE R1,R2,Loop ;			

Instruction Producing Result	Instruction using Result	Latency in clock cycles
FP ALU Op	Another FP ALU Op	3
FP ALU Op	Store double	2
Load double	FP ALU Op	1
Load double	Store double	0

2b	How many bits are in the (0,2) branch predictor with 4K entries? How many entries are in a (2,2) predictor with the same number of bits?	06	4	5
3a	Discuss how consistency is maintained in modern CPUs and GPUs.	08	3	3
3b	Discuss directory-based cache coherence protocol and compare it with snooping protocol.	08	2	2
	OR			
4a	Explain the role of memory fences and barriers in consistency.	08	3	3
4b	Discuss performance issues in DSM systems.	08	3	4
5a	Explain loop unrolling in enhancing loop-level parallelism with example	08	3	3
5b	Analyze the performance trade-offs in vector, SIMD, and GPU architectures.	08	4	4
	OR			
6a	Explain shared, constant, texture, and surface memory usage with examples.	08	3	3
6b	Describe warp divergence and its impact on execution efficiency. Explain methods to reduce it.	08	4	4
7a	For the task graphs given below, determine the following	08	3	3



i. Maximum degree of concurrency.

ii. Critical path length.
iii. Maximum achievable speedup over one process assuming that an arbitrarily large number of processes is available.
iv. The minimum number of processes needed to obtain the maximum possible speedup.
v. The maximum achievable speedup if the number of processes is limited to 2, (b) 4, and (c) 8.

7b	Consider the problem of computing the frequency of a set of itemsets in a transaction database given below. Demonstrate Input, Output, intermediate and both input and output data decompositions for the data given below	08	2	3

Input	Item sets	Frequencies
A,B,C,E,G,H	A,B,C	1
B,D,E,F,K,L	D,E	3
A,B,F,H,L	C,F,G	0
D,E,F,H	A,E	2
F,G,H,K	C,D	1
A,E,F,K,L	D,K	2
B,C,D,G,H,L	B,C,F	0
G,H,L	C,D,K	0
D,E,F,K,L		
F,G,H,L		

OR

8a	Analyse Data Decomposition techniques in detail	10	3	4
8b	Analyse cyclic and block cyclic distributions of mappings based on data partitioning with an example	06	2	4
9a	Explore the differences between the two main types of OpenACC directives you can use to parallelize loops - kernels and parallel using a real code example	10	3	3
9b	With an example code discuss the Parallel directive of OpenAcc	06	3	3

OR

10a	Write an OpenACC program for vector addition	08	4	3
10b	Describe loop-level parallelism in OpenACC using the parallel loop directive. Explain why this approach provides more control to the programmer than the kernels construct.	08	4	4